YONATHAN WONDOSEN

Software Engineer

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with continuous engineering and production experience since 2022, combining rigorous academic computer science training with hands-on professional application development. Proven track record designing, building, and scaling secure, high-performance web applications, enterprise fintech platforms, and healthcare systems. Proficient across **TypeScript (React, Next.js), Node.js (Express), Python (FastAPI, Flask), C#/.NET, REST APIs**, with expertise in **AI-powered RAG pipelines**, microservices optimization, cloud infrastructure (AWS, Azure, VPS), and complex financial system integrations.

WORK EXPERIENCE

Fullstack Software Engineer | VELKAP Ltd

London, UK (Remote) | May 2026 – Present (Contractual)

- Architecting and engineering two distinct enterprise systems for a UK-based fintech firm: an investment management platform and an investor portal backend powered by C#/.NET paired with a Next.js frontend deployed on Azure.
- Designing high-throughput backend APIs for financial transaction processing and investment portfolio tracking.
- Developing highly responsive, accessible user interfaces using modern React patterns, TypeScript, and state management strategies.
- Ensuring strict compliance with UK financial regulatory guidelines and data security standards across all services.
- **Key Achievement:** Engineered high-throughput backend services using .NET, implementing an FCA compliance rules engine, robust OIDC authentication, automated PII redaction, and Azure infrastructure automation for continuous deployment.

Backend Engineer, AI / RAG Development | Akilo Technology

Hong Kong, China (Remote) | June 2025 – September 2025 (Contractual)

- Built a RAG chatbot with two modes: a **public, no-login mode** that collects survey responses through conversation and is **embedded in Qualtrics surveys via custom HTML/JavaScript**, and a **private mode** that answers questions from the client's documents.
- Built pipelines for embedding storage and semantic retrieval with MongoDB vector search (HNSW), grounding LLM answers in the client's content.
- Tuned HNSW parameters, embedding strategy and semantic chunking to improve retrieval relevance and reduce hallucinated answers.
- Deployed the vector search and embedding services in containers on AWS.

Fullstack Software Engineer | EtPay-Rent PLC

Addis Ababa, Ethiopia | March 2025 – November 2025 (Full-time / Hybrid)

- Led full-stack engineering for a real estate and rent collection application, ensuring high availability and system reliability.
- Integrated local and international digital payment gateways, including Chapa/ Telebirr, establishing fault-tolerant transaction workflows.
- Investigated and resolved core payment integration edge cases, directly enhancing payment processing success rates.
- Evaluated third-party integrations and modern architectural paradigms to drive continuous platform optimization.
- **Key Achievement:** Boosted application throughput and reduced response latency by refactoring resource-intensive backend modules and optimizing database query executions.

Co-Founder & Technical Lead | Bego Healthcare Solution (bego.et)

Addis Ababa, Ethiopia | February 2025 – Present

- Built and operate BEGO, an AI-powered EMR for dental and medical clinics, replacing paper-based workflows with digital patient records, clinical workflows and branch analytics (React, Flask).

- Developed a doctor-facing AI assistant that suggests treatment plans for clinician review.
- Selected for the **Ethiopian Artificial Intelligence Institute (EAII) AI Startup Incubation Program**.
- **Key Achievement:** Grew the platform to 5 clinics and 4,000+ patients at 99% uptime, deployed with Docker and PM2 on production VPS infrastructure.

Technology Officer | Ethiopian Kidney Care NGO

Addis Ababa, Ethiopia | April 2024 – November 2026 (Volunteer)

- Managed and maintained the organization's website while integrating external systems to streamline data synchronization and workflows.
- Designed and deployed an automated health screening web application, streamlining patient data validation and campaign analytics.
- Expanded web application capabilities by building user-focused features and responsive user interfaces.
- **Key Achievement:** Built a kidney health screening web app that processed **100+** patients across **1** campaign, replacing manual spreadsheet entry.

SKILLS & COMPETENCIES

Domain	Core Skills & Technologies
Primary Stack	**TypeScript (React, Next.js), Node.js (Express), Python (FastAPI, Flask), C#/.NET, REST APIs**
Secondary Languages & Frameworks	Java Spring Boot, GraphQL
Cloud & DevOps	AWS, Azure, VPS Infrastructure, Docker, PM2, CI/CD Pipelines, Microservices Architecture
AI & Data Engineering	RAG Development, Vector Search (HNSW, MongoDB Vector Search), OpenAI API

Domain	Core Skills & Technologies
Databases	PostgreSQL, MySQL, MongoDB, Prisma ORM
Professional	Project Management, Team Leadership, Technical Documentation, System Architecture

EDUCATION

BSc in Computer Science

Unity University – Addis Ababa, Ethiopia (2022 – 2026)

LANGUAGES

- **English:** Professional Working Proficiency
- **Amharic:** Native
- **Oromo:** Basic Proficiency